Aerospace Information Technology Summer School 2020

Floating Satellite System (FloatSat)

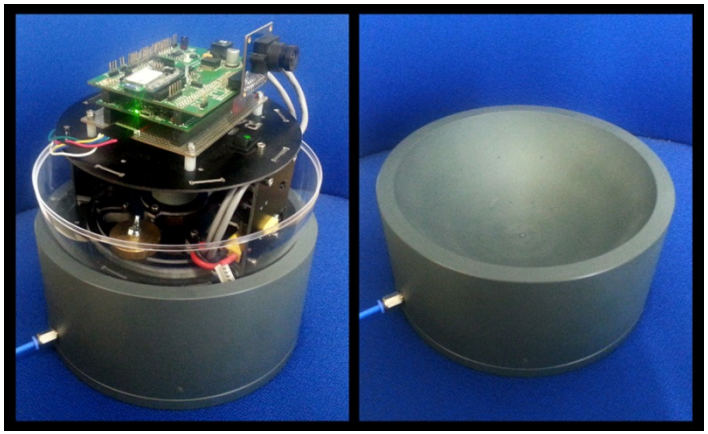
Technical Presentation I

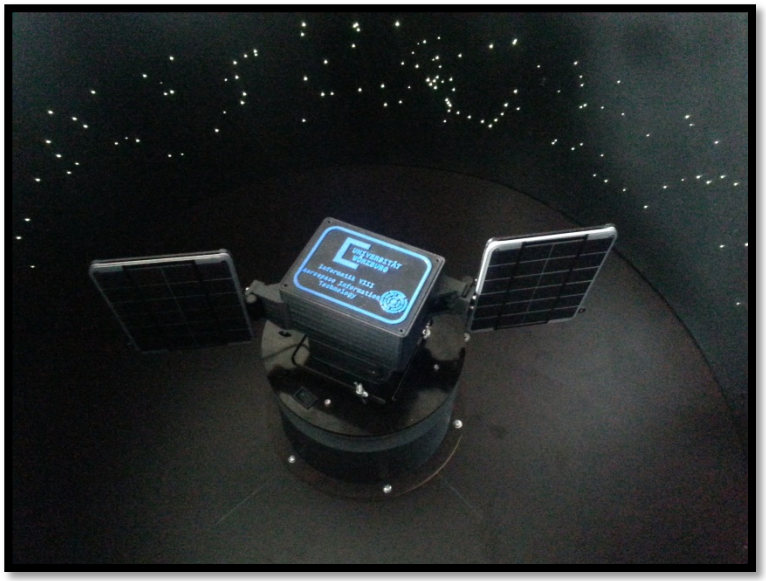
M.Sc. Eng. Atheel Redah

The Floating Satellite (FloatSat) System:

● The Chair of Aerospace Information Technology at the University of Würzburg (Germany) has developed a Floating Satellite (FloatSat) system in order to help students understand and get familiar with basic satellite subsystems and also to develop and test different attitude control algorithms and strategies for small satellites in an almost frictionless environment similar to that in space.

● The FloatSat system is consists mainly from a mechanical structure that contains the basic satellite subsystems. This structure is placed into a hemisphere shell that it is floating inside a Spherical Air Bearing unit. The air bearing unit requires compressed air input may vary depending on the mass of the floating unit. In order to monitor and command the FloatSat, a ground station computer could be used to communicate via the Bluetooth or the Wi-Fi module available in the satellite through commanding and telemetry display tool.

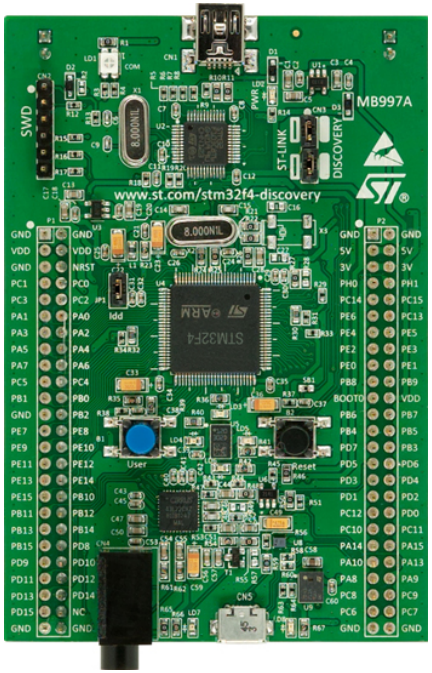




Floating Satellite System (FloatSat)

Embedded System

The STM32F4DISCOVERY Development Board:



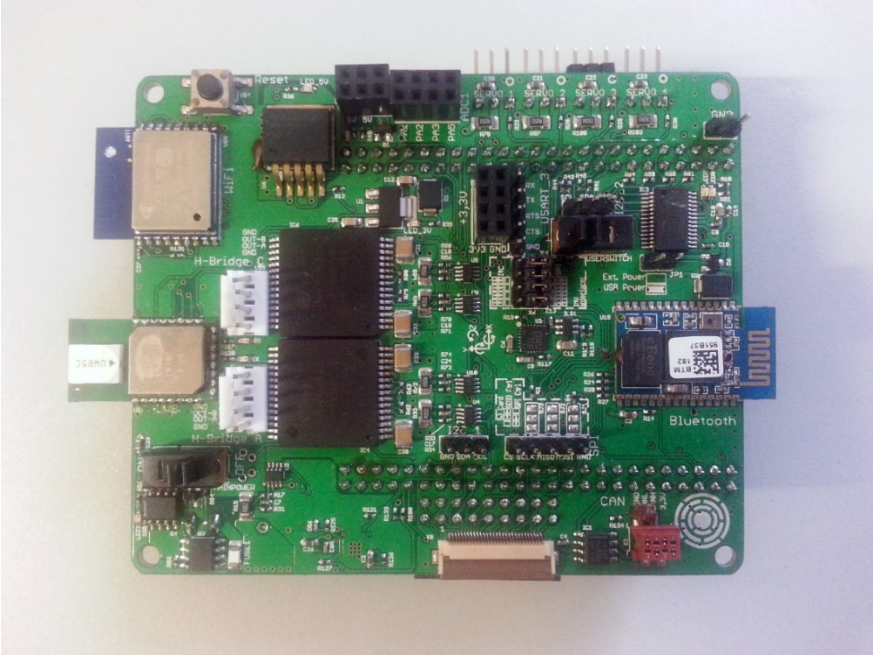
- **STMicroelectronics** – www.st.com.
- **Programmer:** On-board ST-LINK/V2 in-circuit debugger and programmer.
- **Board power supply**: Through USB bus or from an external 5 V supply voltage.
- **Core:** ARM 32-bit Cortex™-M4 CPU with FPU, ART Accelerator™ (Adaptive real-time accelerator) allowing 0-wait state execution from Flash memory, frequency up to 168 MHz, MPU (Memory Protection Unit), 210 DMIPS/1.25 DMIPS/MHz (Dhrystone 2.1), and DSP instructions.
- **Memories:** 1 MB of Flash memory (Application), 192+4 KB of SRAM including 64KB of CCM (Core Coupled Memory) data RAM (Working).
- **3×12-bit A/D converters:** up to 24 channels with 2.4 MSPS and 7.2 MSPS in triple interleaved mode.
- **2×12-bit D/A converters**.
- **General-purpose DMA:** 16-stream DMA controller with FIFOs and burst support.
- **17 x Timers:** up to 12x16-bit and 2x32-bit timers up to 168 MHz, each with up to 4 IC/OC/PWM or pulse counter and quadrature (incremental) encoder input.
- **82 x GPIO pins:** with interrupt capability.
- **15 communication interfaces:** 3×I2C interfaces (SMBus/PMBus), 4xUSARTs / 2xUARTs (10.5 Mbit/s, ISO 7816 interface, LIN, IrDA, modem control), 3xSPIs (42 Mbits/s), 2×CAN interfaces (2.0B Active), SDIO interface.
- **Advanced connectivity:** USB 2.0 full-speed device/host/OTG controller with on-chip PHY, USB 2.0 high-speed/full-speed device/host/OTG controller with dedicated DMA, on-chip full-speed PHY and ULPI–10/100 Ethernet MAC with dedicated DMA which supports IEEE 1588v2 hardware, MII/RMII.
- **8- to 14-bit parallel DCMI** (Digital Camera Interface) up to 54 Mbytes/s, and parallel **LCD interface** (8080/6800 modes).
- True **RNG** (Random Number Generator) and **CRC** (Cyclic Redundancy Check) calculation unit.

The STM32F4DISCOVERY Extension Board V3.0:

- **University of Würzburg, Chair of Computer Science VIII**
- **Board power supply:** through **USB** bus , or from an external **10.5 to 20 V** supply voltage.
- **9- or 12-Volt onboard DC-DC Converter.**
- **3.3- and 5-Volt onboard voltage regulators**.
- **Deep Discharge Battery Protection**.
- **Onboard Modules:**

- BTM-182 Bluetooth <==> UART2 (STM32F4).
- ESP-WROOM-02 WiFi <==> UART3 (STM32F4).
- DWM1000 Ultra Wide Band Transceiver <==> SPI2 (STM32F4).
- LSM9DS0 Inertial Measurement Unit (IMU) <==> I2C2 (STM32F4).
- FT232RL USB-to-UART Bridge <==> UART3 (STM32F4).
- 4x VNH3SP30 H-Bridge Motor Driver <==> TIM4 (STM32F4).
- 9x INA219A Current Monitor <==> I2C1 (STM32F4).

- **Camera Interface <==> DCMI + I2C1 (STM32F4).**
- **SD/MMC Card Interface <==> SDIO** or **SPI1 (STM32F4).**
- **CAN Interface <==> CAN1 (STM32F4).**
- **Pinout Interface:**



ADC		TIM		I2C		UART		SPI		GPIO
ADC123_IN1	PA1	TIM1_CH1	PE9	I2C1_SCL	PB8	USART2_TX	PD5	SPI1_CS	PA15	PD12 [LED Green]
ADC123_IN2	PA2	TIM1_CH2	PE11	I2C1_SDA	PB9	USART2_RX	PD6	SPI1_SCK	PB3	PD13 [LED Orange]
ADC123_IN3	PA3	TIM1_CH3	PE13	I2C2_SCL	PB10	USART2_CTS	PD3	SPI1_MISO	PB4	PD14 [LED Red]
ADC12_IN5	PA5	TIM1_CH3	PE14	I2C2_SDA	PB11	USART2_RTS	PD4	SPI1_MOSI	PB5	PD15 [LED Blue]
		TIM4_CH1	PD12			USART3_TX	PD8	SPI2_CS	PB12	
		TIM4_CH2	PD13			USART3_RX	PD9	SPI2_SCK	PB13	
		TIM4_CH3	PD14			USART3_CTS	PD11	SPI2_MISO	PC2	
		TIM4_CH3	PD15			USART3_RTS	PB14	SPI2_MOSI	PB15	

What do we need to start programming the STM32F4?

● **IDE (Integrated Development Environment) for C/C++ Developers**: An IDE is a software application that provides comprehensive facilities to computer programmers for software development. An IDE normally consists of a **source code editor**, **build automation tools** and a **debugger**.

● **ARM GNU Toolchain**: An ARM GNU toolchain is a complete development environment for embedded C/C++ development on ARM. The toolchain normally consists of **GNU C and C++ compilers**, **GNU assembler & linker**, **C and C++ runtime libraries** and **GNU debugger**.

● **ST-LINK Utility** software for fast in-system programming (ISP) of the STM32 microcontroller via the on-board **ST-LINK/V2**.

● **GDB server for debugging.**

● **Real Time Operating System Project Library configured to use the Firmware Library of the STM32F4 and a Template Project linked with the library.**

RODOS (Realtime Onboard Dependable Operating System):

- Preemptive Real Time Operating System
- Open Source Project
- Developers:
  - German Aerospace Center (DLR)
  - Department for Aerospace Information Technology at the University of Würzburg.
- C++ API
- Easy to schedule tasks
- Easy to share tasks
- Middleware with Topics and Subscribers
- Gateway brings Topics to other RODOS nodes

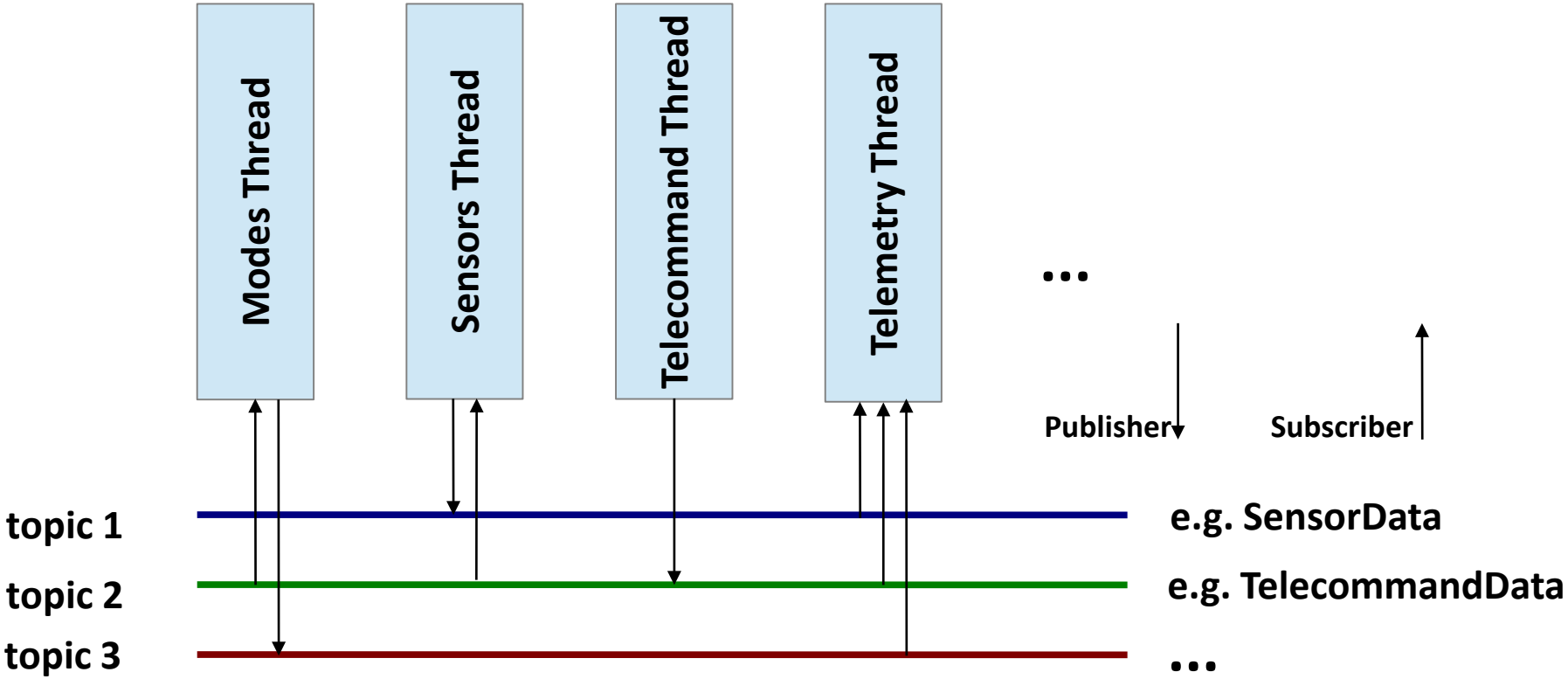
```
class Telemetry: public Thread {

public:

Telemetry(const char* name) : Thread(name) {}

void init() {GreenLED.init(true, 1, 0);}

void run() {
            while (1) {
                        GreenLED.setPins(~GreenLED.readPins());
                        PRINTF("Hello Rodos, the time now is %f \r\n",SECONDS_NOW());
                        suspendCallerUntil(NOW()+1000*MILLISECONDS);}
            }
};
Telemetry Telemetry("Telemetry");
```

Time NOW() in seconds as double

Returns the time now in the internal time resolution (eg. Nanoseconds).

Suspend the thread until a given time or a wake up by a call to the method resume.

FloatSat:


Modes Thread
Sensors Thread
Telecommand Thread
Telemetry Thread
...
Publisher
Subscriber
topic 1
topic 2
topic 3
e.g. SensorData
e.g. TelecommandData
...

Floating Satellite System (FloatSat)

Modules & Sensors

Inertial Measurement Unit (IMU):

● Inertial Measurement Unit (IMU), is an electronic device that measures and reports on an object's velocity, orientation, and gravitational forces, using a combination of accelerometers, gyroscopes, and magnetometers.

● The LSM9DS0 is an inertial measurement unit (IMU) that packs an 3-axis Gyroscope with 3-axis Accelerometer and 3-axis Magnetometer:

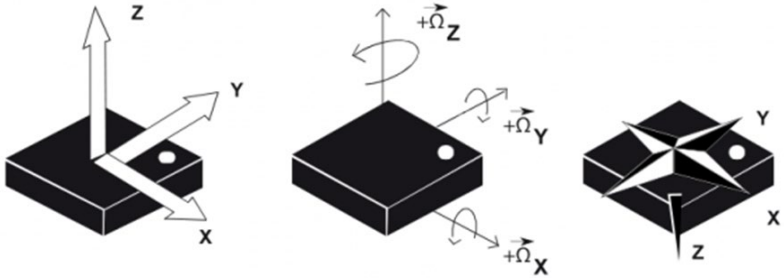
* SPI / I2C serial interfaces.

* 16-bit data output per axis.

* Sensitivity range (configurable):
- Gyro: ±245, ±500, or ±2000 dps angular rate full scale.
- Accelerometer: ±2, ±4, ±6, ±8 or ±16 g linear acceleration full scale.
- Magnetometer: ±2, ±4, ±8, or ±12 gauss magnetic full scale.

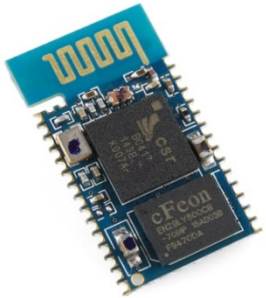
* Embedded temperature sensor.

● Orientation Angles:
- Three attitude angles are referenced to the local horizontal plane which is perpendicular to earth's gravity:
* Yaw is defined as the angle between the sensor X-axis and the magnetic north on the horizontal plane.
* Pitch is defined as the angle between the sensor X-axis and the horizontal plane.
* Roll is defined as the angle between the sensor Y-axis and the horizontal plane.



Wireless Modules:

- **Bluetooth Module BTM-182:**
  - Class 2 module with PCB antenna
  - SPP firmware
  - Configurable with AT commands
  - Bluetooth standard Ver. 2.0 + EDR compliant.
  - Low current consumption
  - Hold, Sniff, Park, Deep sleep modes
  - 3.0V or 1.8V operation
  - Support for up to seven slaves : SCO links <3>, ACL links, Piconet <7>
  - Interface: USB, UART, and PCM (requires voice CODEC)
  - Support for 802.11 co-existence



- **WiFi Module ESP-WROOM-02:**
  - IEEE 802.11b/g/n at 2.4 GHz
  - Integrated low power 32-bit MCU
  - Integrated 10-bit ADC
  - Integrated TCP/IP protocol stack
  - Integrated TR switch, balun, LNA, power amplifier and matching network
  - Integrated PLL, regulators, and power management units
  - Various interfaces available, including SDIO,HSPI,UART,I2C,I2S,IR,PWM,GPIO, etc.
  - Deep sleep power <10uA, Power down leakage current < 5uA
  - Wake up and transmit packets in < 2ms
  - Standby power consumption of < 1.0mW +20dBm output power in 802.11b mode

Wireless Modules:

- **UWB Module DWM1000:**

- IEEE802.15.4-2011 Ultra-Wide Band compliant wireless transceiver module

- Enables the location of objects in real time location systems (RTLS) to a precision of 10cm indoors

- High data rate communications up to 6.8 Mb/s

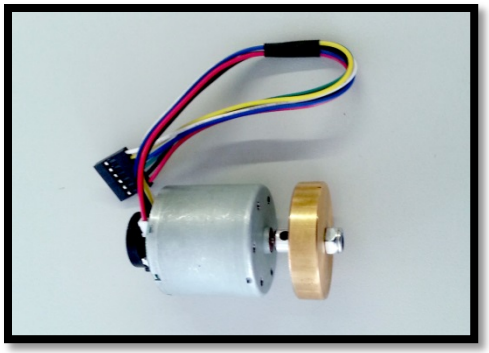
- Excellent communications range of up to 300m

- High immunity to multipath and interference

Floating Satellite System (FloatSat)

Actuators

Reaction Wheel:



● A reaction wheel or a momentum wheel in a spacecraft is an electric motor attached to a flywheel.

● When the wheel rotation speed is changed, a torque is generated:

$$\tau = J_w \frac{d\omega_w}{dt}$$

Where J_w is the wheel moment of inertia and ω_w is the wheel rotation speed.

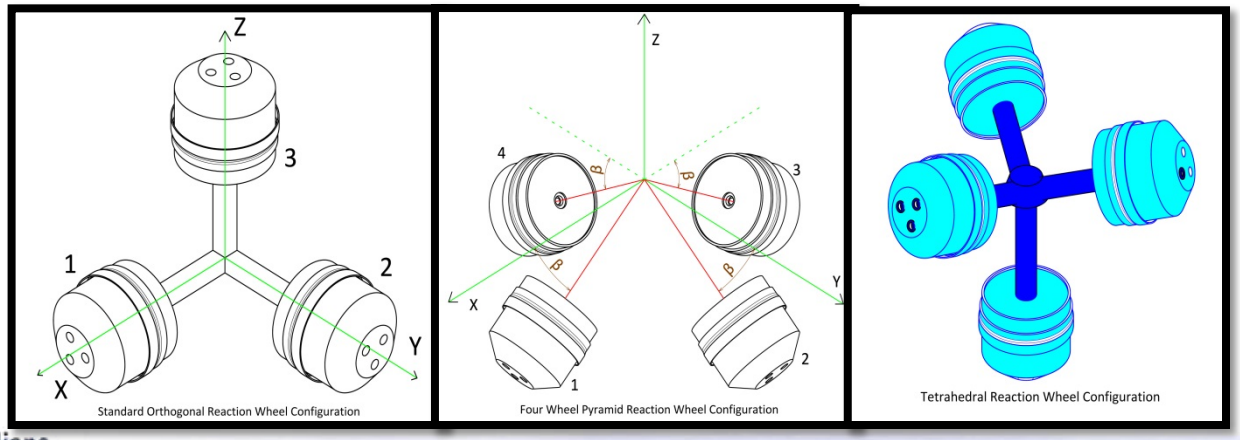
● According to conservation of angular momentum law, the generated torque will cause the spacecraft to counter rotate proportionally around its center of mass in the same plane of rotation:

$$\tau_s = -J_w \frac{d\omega_w}{dt} \rightarrow J_s \frac{d\omega_s}{dt} = -J_w \frac{d\omega_w}{dt}$$

$$\omega_s = -\frac{J_w}{J_s} * \omega_w \; and \; \theta_s = -\frac{J_w}{J_s} * \int_0^t \omega_w . d\tau$$

Where J_s is the spacecraft moment of inertia, ω_s is the spacecraft rotation speed and θ_s is the spacecraft angular position.

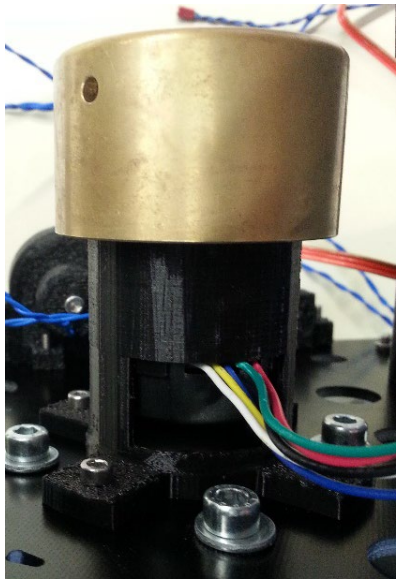
● For precision three axis attitude control of a spacecraft, at least three reaction wheels are needed:

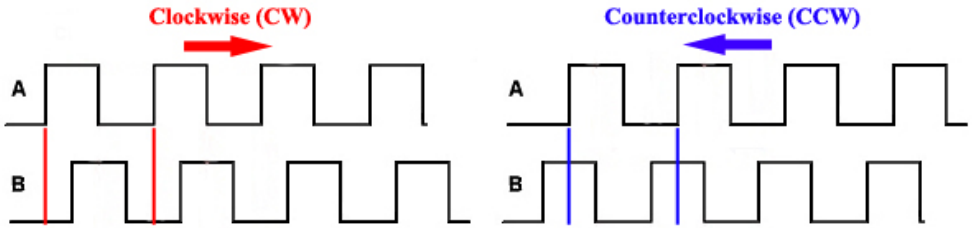


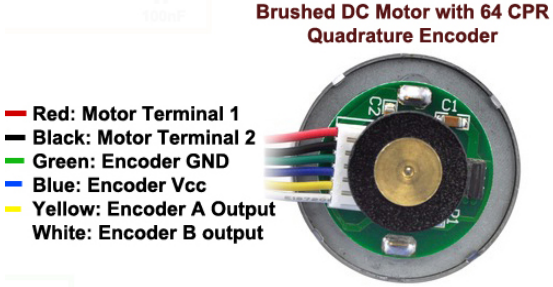
Actuator & Power Driver:

Motorized Reaction Wheel:

- **Pololu Brushed Motor with 64 CPR Encoder**
- **1-12V** Operating voltage.
- Key specs at **12 V**: **11,000 RPM** and **300 mA** free-run, **0.3 kg-cm** and **5 A stall**.
- Integrated **64 CPR** (counts per revolution) quadrature **Hall effect encoder**.
- **Reaction Wheel**: **298 g @ 1.175*10-4 Kgm2**
- **The quadrature encoder has two square wave outputs (A and B) which are 90 degrees out of phase with each other**



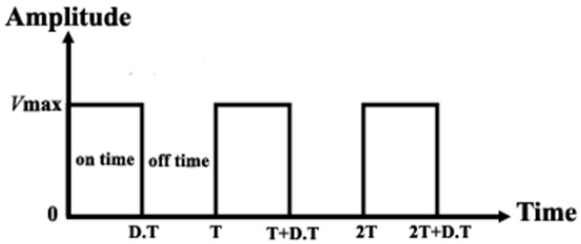




Pulse Width Modulation (PWM):

● Pulse Width Modulation (PWM) is a powerful technique for controlling analog systems that expects an analog signal of varying amplitude with microcontroller's digital outputs:

* The voltage or current source is supplied to the analog load by means of repeating series of on and off pulses.
* The on-time is the time during which the DC supply is applied to the load.
* The off-time is the times during which that supply is switched off.
* The term duty cycle D describes the proportion of on-time to the regular interval or period of time T for the PWM signal. Duty cycle is expressed in percent, where 100% being fully on and 0% being fully off.
* PWM controlled drivers regulate the effective applied voltage by changing the duty ratio of on-time to off-time at a given frequency.



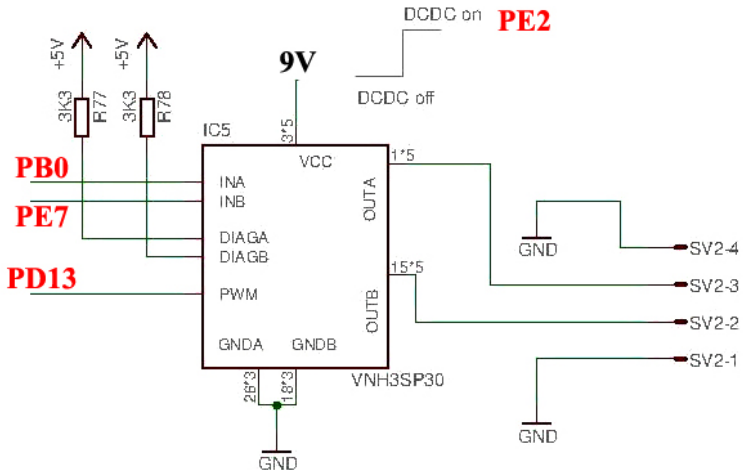
* If the pulse waveform has a low value equal to zero, a high value equal to $V_{\max}$ and a duty cycle D, then the average value of the waveform V_{ave} can be given by following equation :

$$V_{ave} = \frac{1}{\mathrm{T}} \int_0^{\mathrm{DT}} V_{\max} \cdot dt = \mathrm{D} V_{\max}$$

Actuator & Power Driver:

Power Driver:

- **ST VNH3SP30** 30A Motor Driver (**H-Bridge**)
- Minimum operating voltage: **5.5 V**
- Maximum operating voltage: **16 V**
- Continuous output current per channel: **9 A**
- Peak output current per channel: **30 A**
- Maximum PWM frequency: **10 kHz**
- Features various fault protection features such as over-temperature and over-current shutdown
- **Has two input control signals (INA, INB), which can choose one of four modes of operation: Clockwise CW (1, 0), Counterclockwise CCW (0, 1), short brake (1, 1), and stop mode (0, 0).**



H-Bridge B

Power Source:

2x 2100mAh 2S1P 20C LiFePo4 Battery Pack:

- Capacity: 2100mAh
- Voltage: 2S1P / 2 Cell / 6.6V (7.2V fully charged)
- Discharge: 20C Constant / 40C Burst
- Weight: 103g (including wire, plug & case)
- Dimensions: 80 x 44 x 16mm
- Balance Plug: JST-XH
- Discharge Plug: JR & Futaba

Floating Satellite System (FloatSat)

Operation Modes & Requirements

Ground Station GUI & Operation Modes:

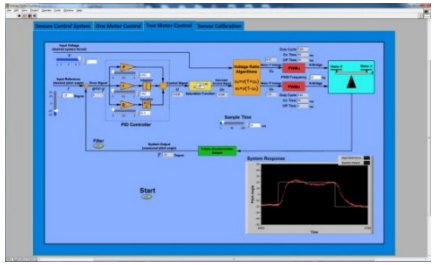
What are the requirements?

- Implementing the Telecommand and the Telemetry communication protocols for communication with the satellite
- Implementing an easy to use user interface where all the necessary information can be gathered from the satellite and displayed on the GUI using charts, numerical, string and LED indicators.

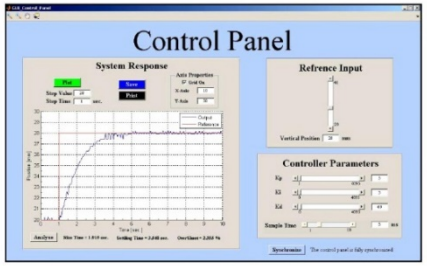
Operation Modes:

- **Standby Mode.**
- **Motor Speed Mode:** CW and CCW **rotation** with **desired motor rotation speed** — **Controller 1**
- **Velocity Mode:** CW and CCW **rotation** with **desired angular velocity** — **Controller 2**
- **Position Mode: Stabilizing** at **desired orientation** (Compass Angle) — **Controller 3**
- **Mission Mode**

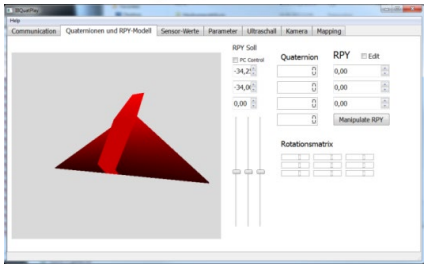
LabView, Matlab, QT or …?



Graphical Programming



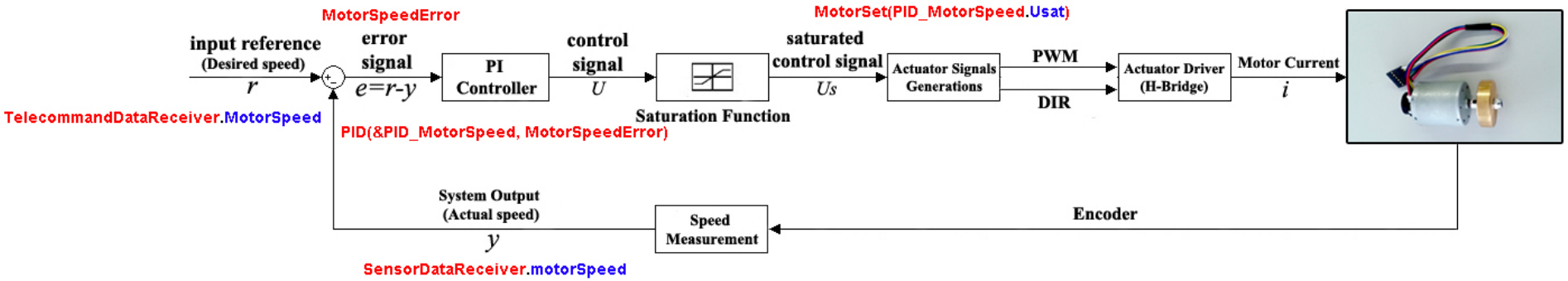
M-Code



C++

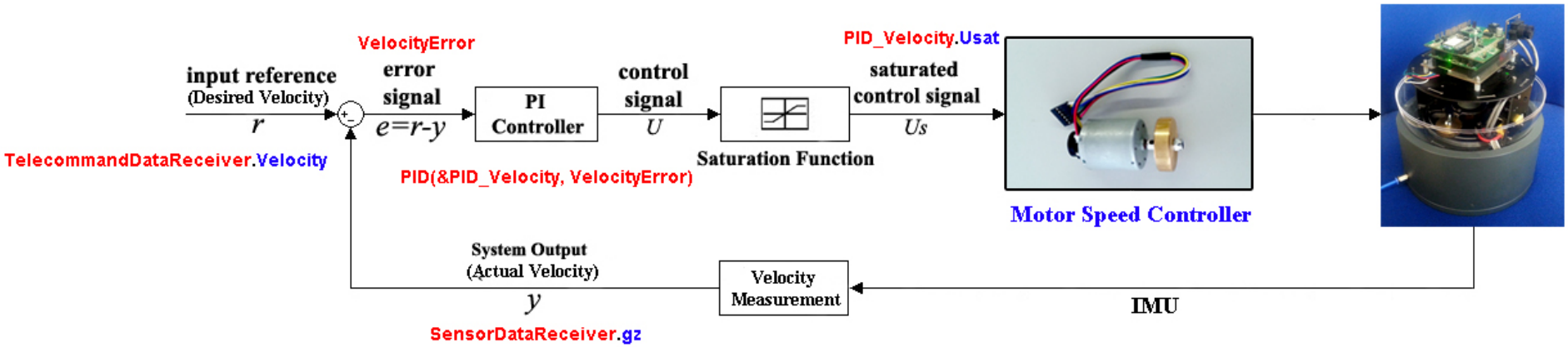
Motor Speed Mode:

- A PI controller for controlling the Motor Rotation Speed needs to be implemented as following:



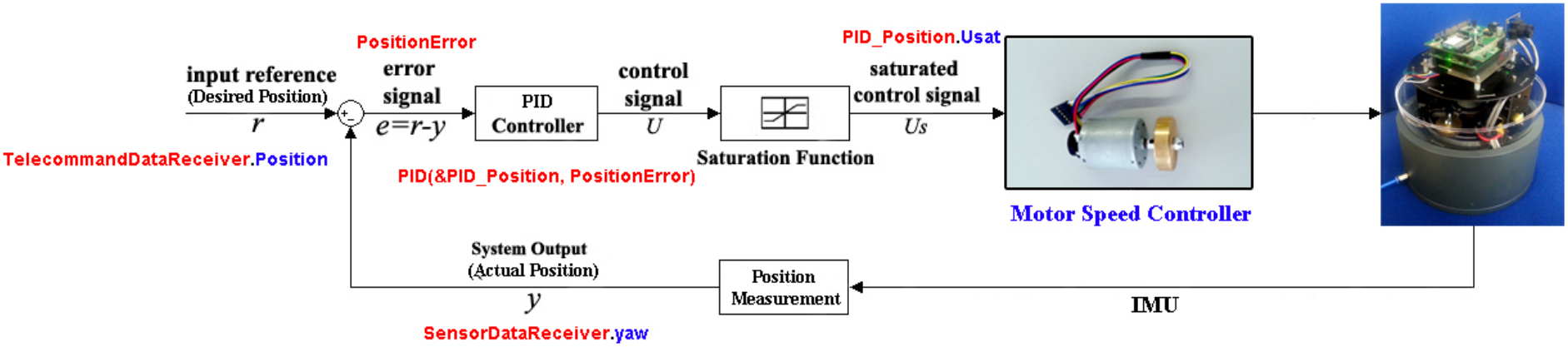
Satellite Velocity Mode:

- A PI controller for controlling the FloatSat Angular Velocity needs to be implemented as following:



Satellite Position Mode:

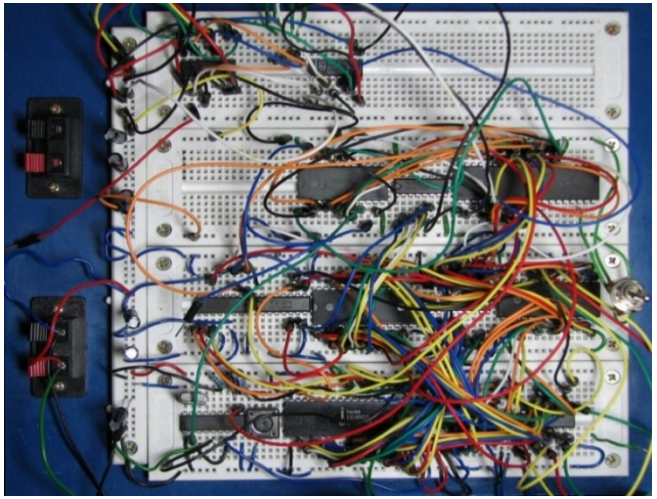
- A PID controller for controlling the FloatSat Angular Position needs to be implemented as following:



Electrical and Power System Implementation:

What are the requirements?

- Battery Power measurements.
- Very good wiring alignments.
- If necessary, adding noise suppression elements to the motor.
- Minimize the damage you may cause to different components during operation.
- Real time power consumption measurement to calculate the expected life time of the satellite in order to feed these information back to the ground station.



Mechanical Design & Construction:

What are the components need to be include in my design?

- **Power Source**: **Two battery Packs and solar panels**.
- **Development Board** and **Extension** Board.
- **Motorized Reaction Wheel.**
- **Masses for CoM Adjustment.**
- **Extra Electronic Components Board**.
- **Payload** for the **Satellite Mission.**

How can I distribute the components in order to keep the center of mass alonge the vertical axis of the bottom half of the sphere?

What are the available materials and tools so my design will be feasible?

How can I make my design more attractive and flexible with easy access to the batteries and other components?